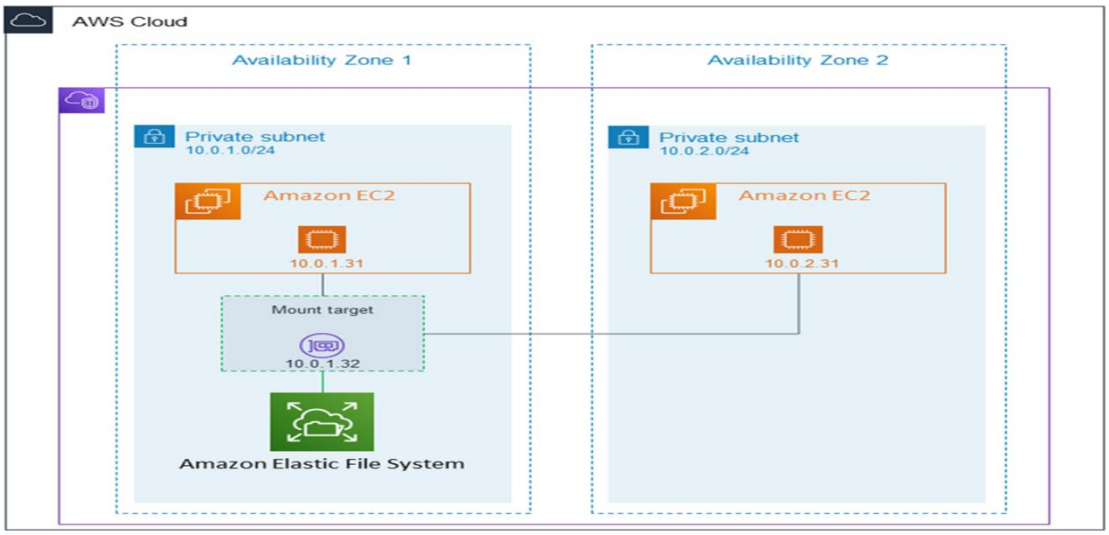
Elastic File Storage

Amazon Elastic File System (EFS) is a scalable and fully managed file storage service provided by Amazon Web Services (AWS). It allows you to create and configure file systems that can be accessed concurrently from multiple EC2 instances, providing a highly available and scalable storage solution for your applications and workloads.

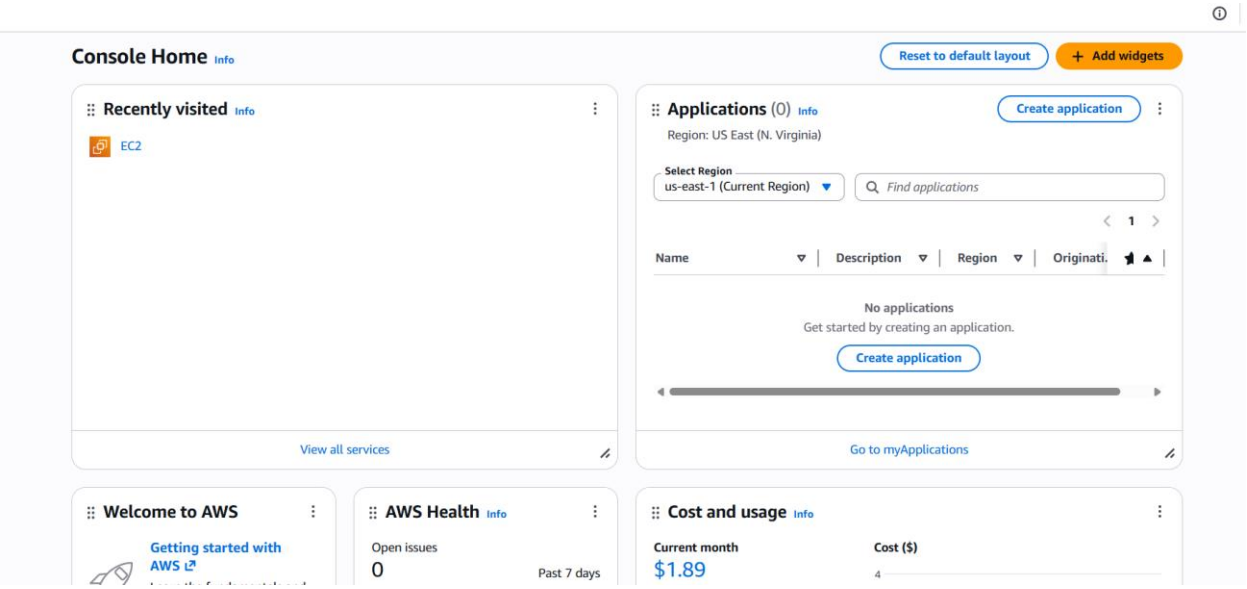


Amazon EFS supports **for only Amazon Linux instance**:

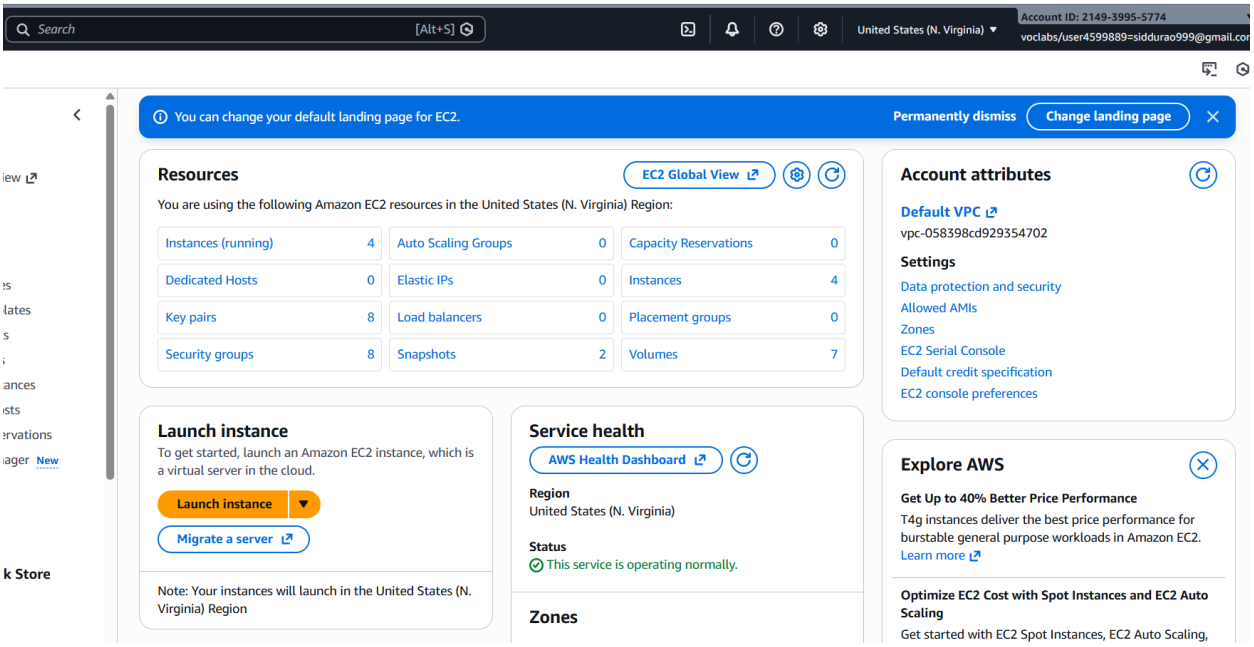
Step 1: Launching EC2 Instances

1. Sign in to AWS Management Console.

2. Go to the EC2 dashboard.



3. Click on "Launch Instance".



4. Configure instance details:

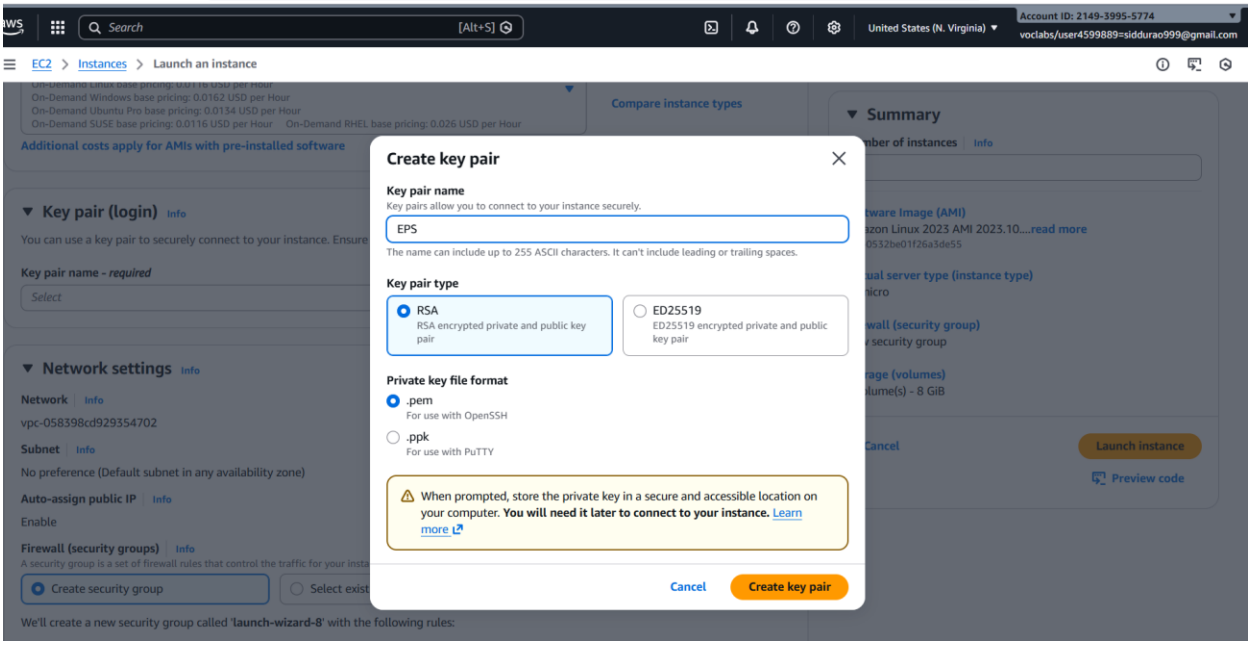
- Instance Name: **EFS-1**

4. Choose an Amazon Machine Image (AMI) **for Linux**.

5.

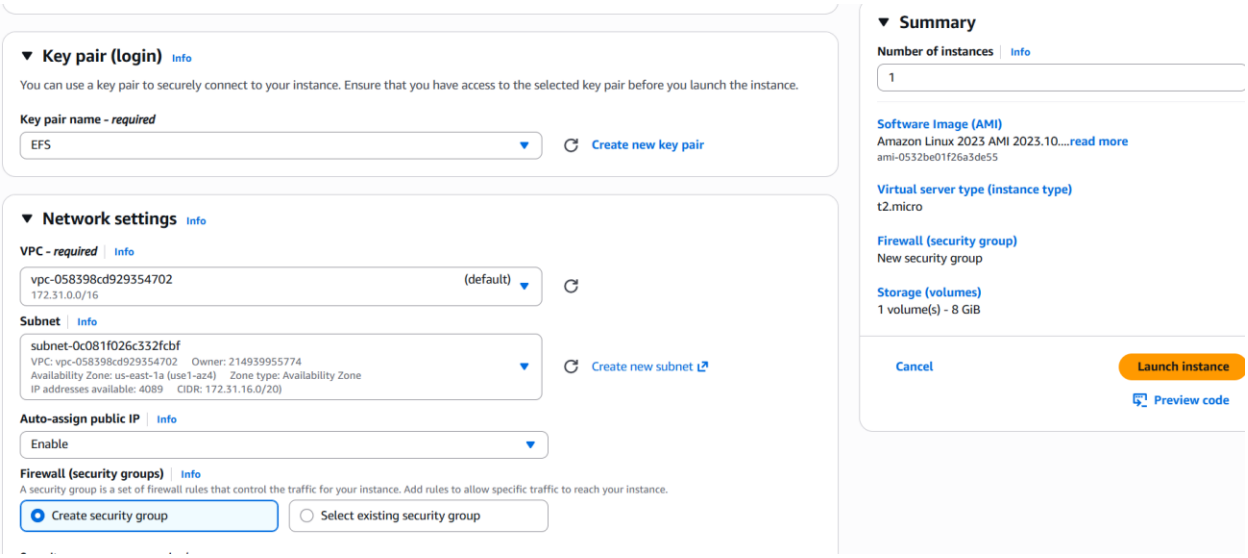
6. Select instance type: t2.micro.

- Key Pair: EFS

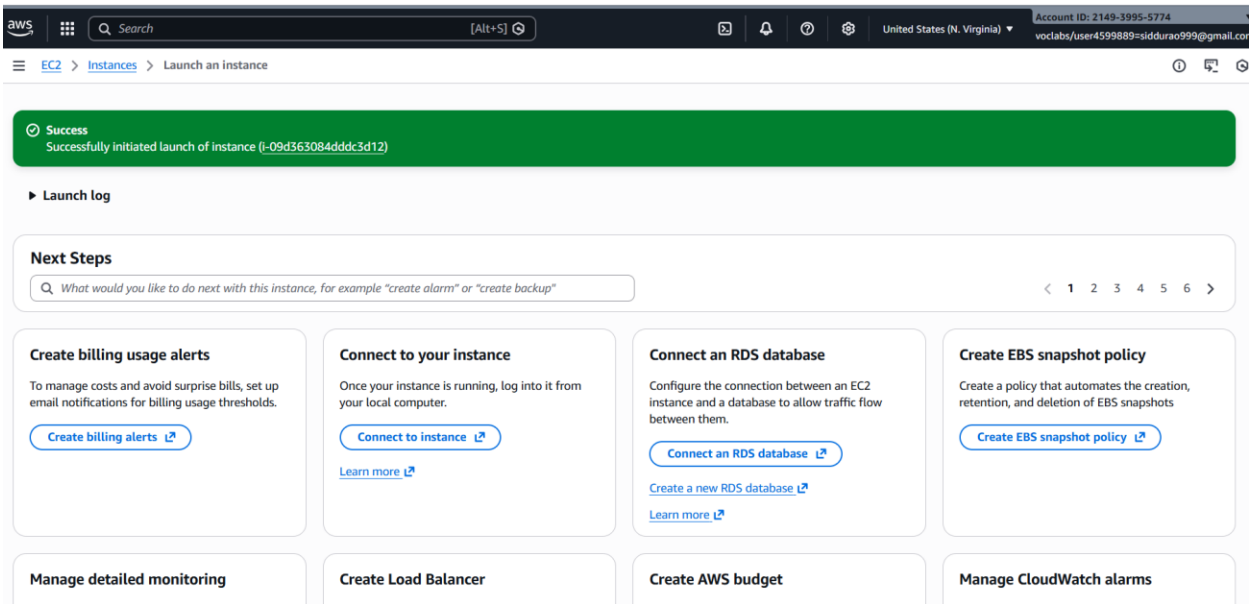


- Network: Choose Subnet-1a

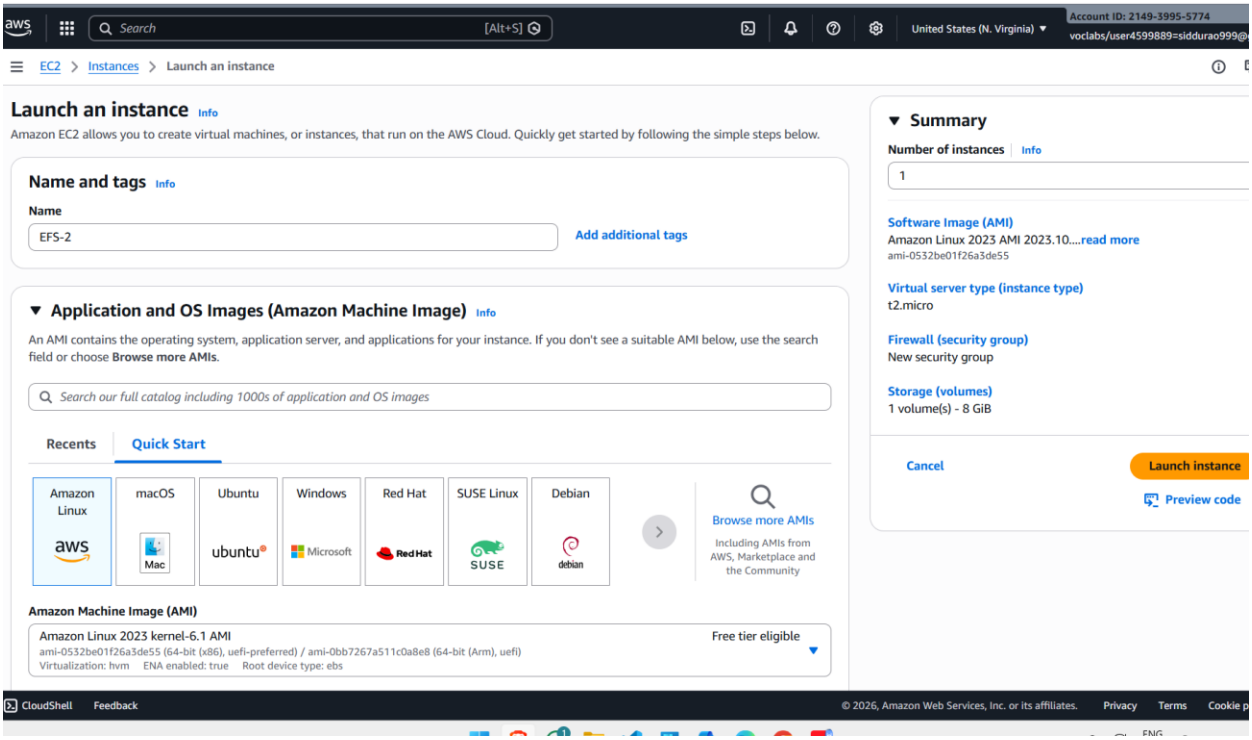
- Security Group: **Create a new security group (SG) and add NFS and allow from anywhere.**



7.Launchtheinstance.



8. Repeat the above steps to launch another instance named **EFS-2 in Subnet-1b** with the same configuration.



Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

EPS-2 | Create new key pair

Network settings Info

VPC - *required* | Info

vpc-058398cd929354702 (default)
172.31.0.0/16

Subnet | Info

subnet-0bde714fa85eb4e22
VPC: vpc-058398cd929354702 Owner: 214939955774
Availability Zone: us-east-1b (use1-az6) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.32.0/20)

Create new subnet

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

Security group name - *required*

launch-wizard-9

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-

Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0532be01f26a3de55

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Launch instance | Preview code

EC2 > Instances > Launch an instance

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

Security group name - *required*

launch-wizard-9

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* | Info

launch-wizard-9 created 2026-02-06T04:31:54.750Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 2049, 0.0.0.0/0) | Remove

Type	Protocol	Port range
NFS	TCP	2049

Source type	Source	Description - *optional*
Custom	Add CIDR, prefix list or security group 0.0.0.0/0 ✕	e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Add security group rule

Summary

Number of instances | Info

1

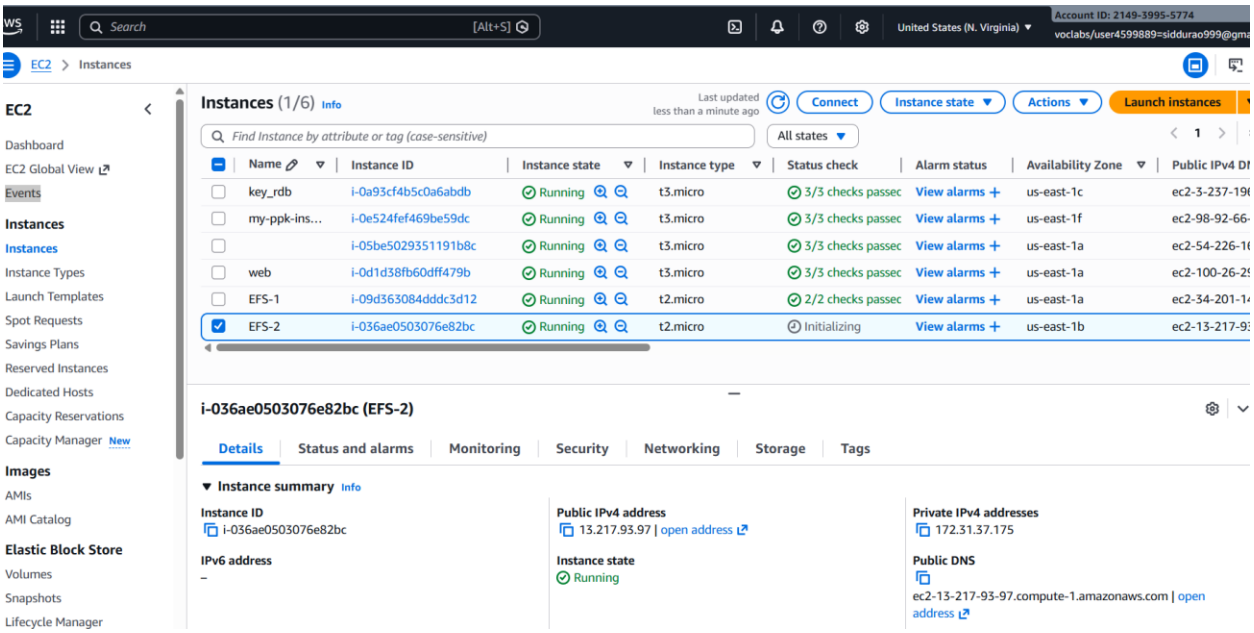
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read mo
ami-0532be01f26a3de55

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

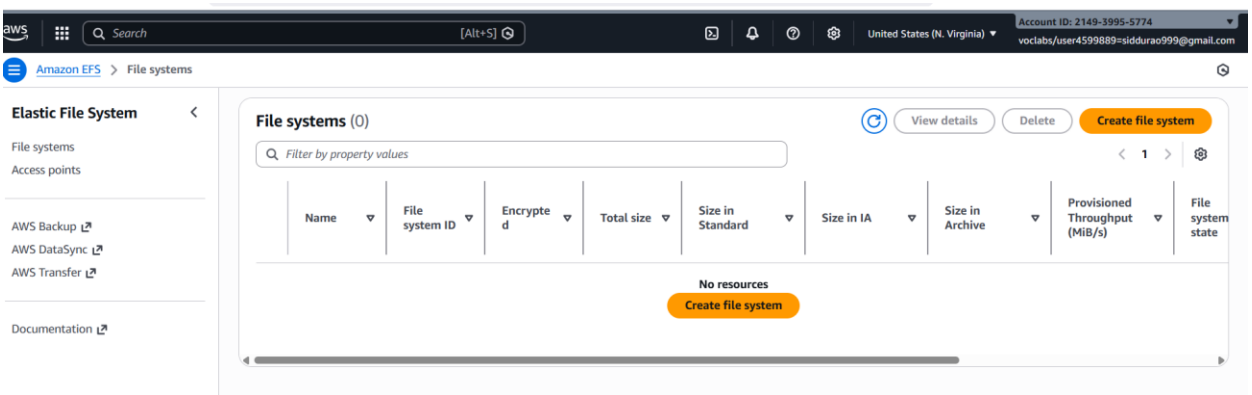
Storage (volumes)
1 volume(s) - 8 GiB

Cancel



Step 2: Creating an EFS File System

1. Go to the EFS service in the AWS Management Console.



Click on "Create file system".

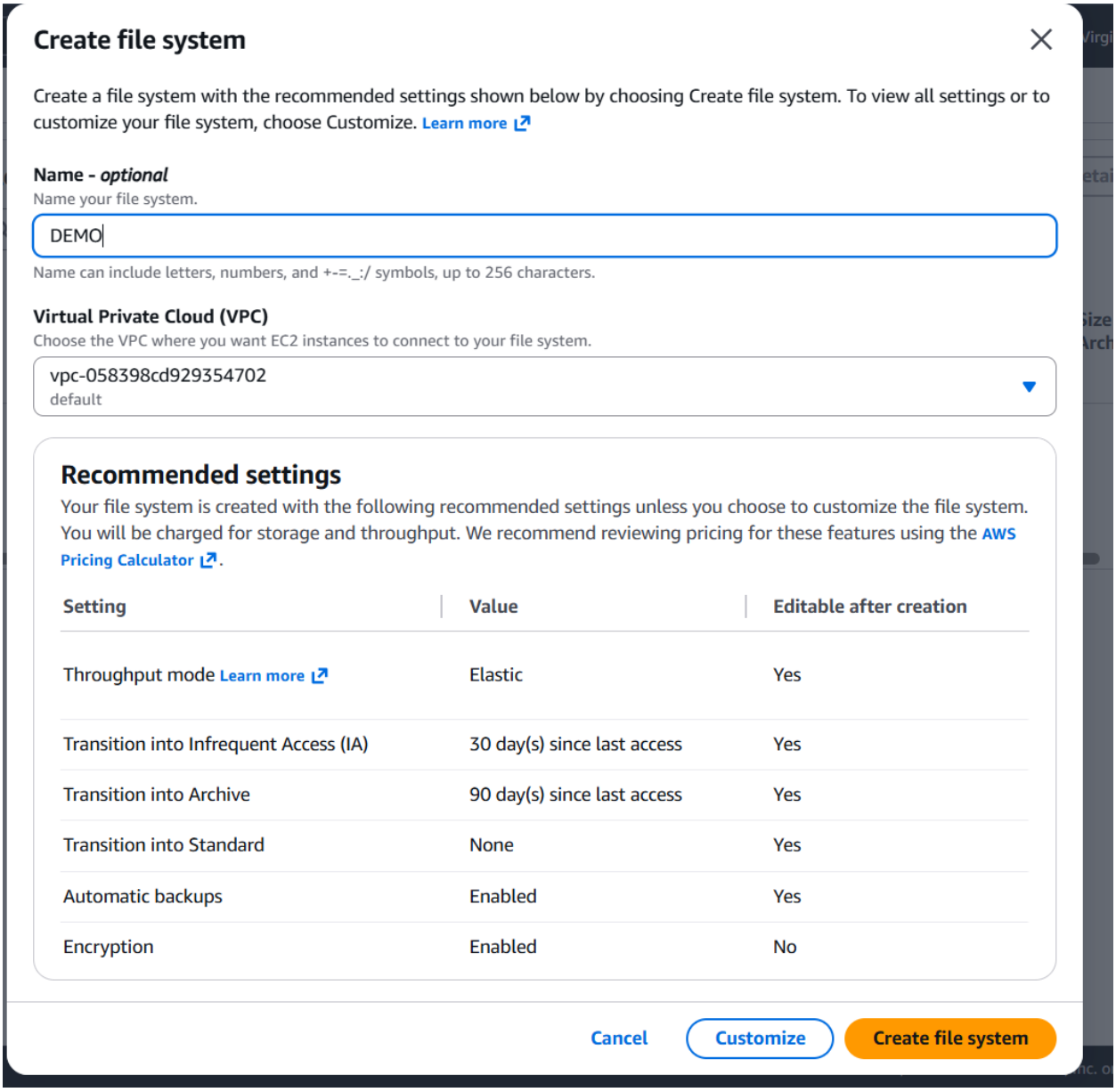
3. Specify details:

- Name: Optional

- VPC: Default

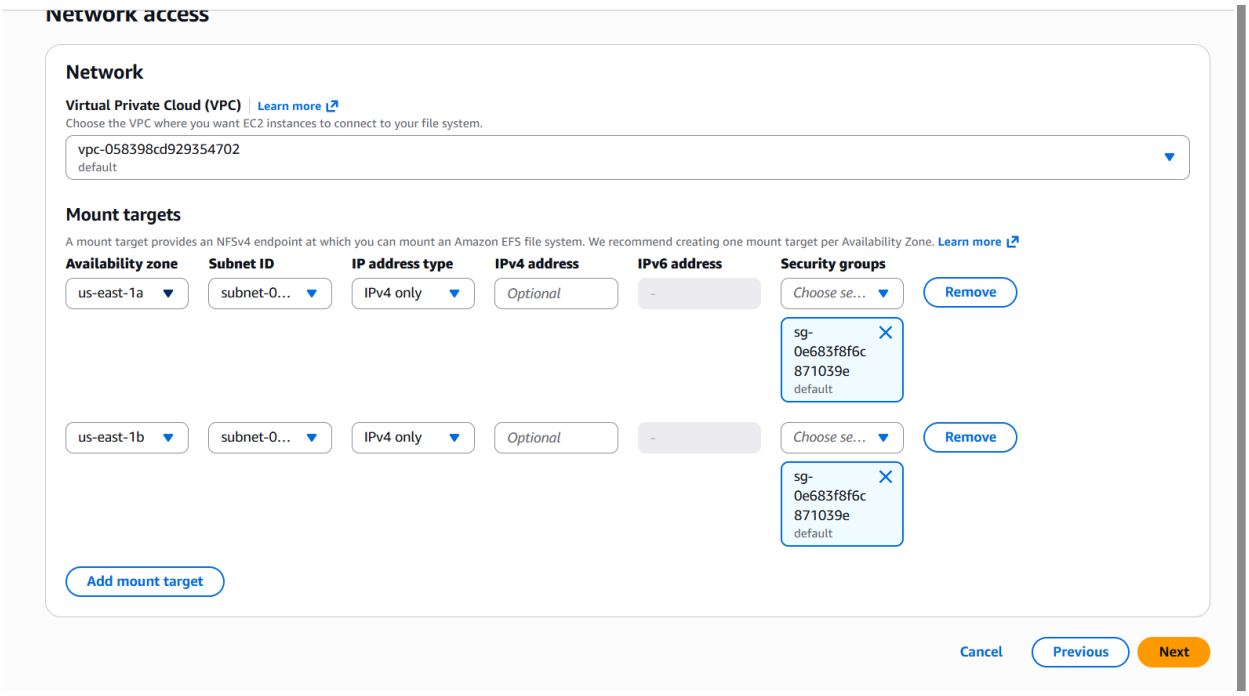
- Enable region button.

Click on "Next".



4. In the Network settings:

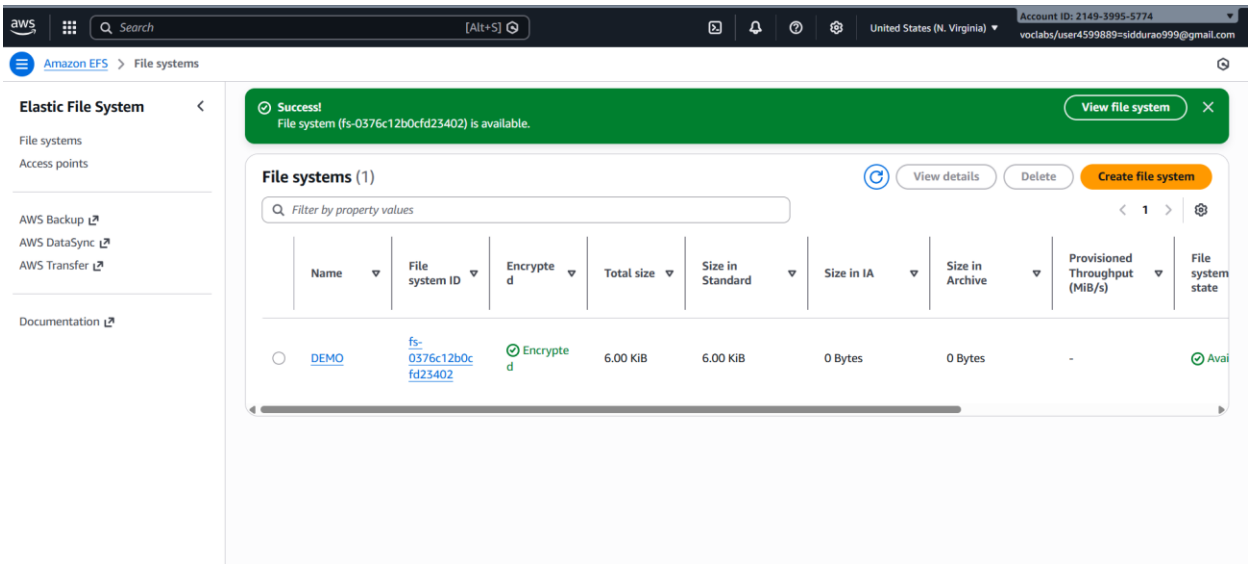
- Delete all security groups.



- **Select the newly created security group (**NFS).---which is created while creating instance

Click on "Next" and review the configuration.

Click on "Create file system".



Step 3: Accessing the two EC2 instances named EFS-1 & EFS -2 in **two different PowerShell** sessions and performing the specified tasks:

Accessing EFS-1 Instances in Two Different PowerShell Sessions:

1. Open Two PowerShell Sessions:

- Open two separate PowerShell windows or tabs on your local machine.

For each Instance:

2. SSH into the Instance:

- Use the SSH command to connect to the EFS-1 instance

ssh -i [path-to-your-keypair.pem] ec2-user@[instance-public-ip]

```
PS C:\Users\DELL\downloads> ssh -i "EFS.pem" ec2-user@ec2-18-208-129-16.compute-1.amazonaws.com
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-16-4 ~]$
```

```
PS C:\Users\vivek vardhan\downloads> ssh -i "EPS.pem" ec2-user@ec2-52-204-114-222.compute-1.amazonaws.com
The authenticity of host 'ec2-52-204-114-222.compute-1.amazonaws.com (52.204.114.222)' can't be established.
ED25519 key fingerprint is SHA256:t0AJsL1Bm3hCgxuoJ8cocK2fTl2L/YWcguANccmT1vY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-52-204-114-222.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-31-141 ~]$
```

3. Switch to Root User:

- Gain root access by executing the following command:

sudo su

4. Create a Directory:

- Make a directory named "efs" using the following command:

mkdir efs

```
[ec2-user@ip-172-31-16-4 ~]$ sudo su
[root@ip-172-31-16-4 ec2-user]# mkdir efs
[root@ip-172-31-16-4 ec2-user]# ls
efs
[root@ip-172-31-16-4 ec2-user]#
```

5. Install Amazon EFS Utilities:

- Use yum package manager to install the Amazon EFS utilities:

yum install -y amazon-efs-utils

```
[root@ip-172-31-16-4 ec2-user]# yum install -y amazon-efs-utils
Amazon Linux 2023 Kernel Livepatch repository                                                
```

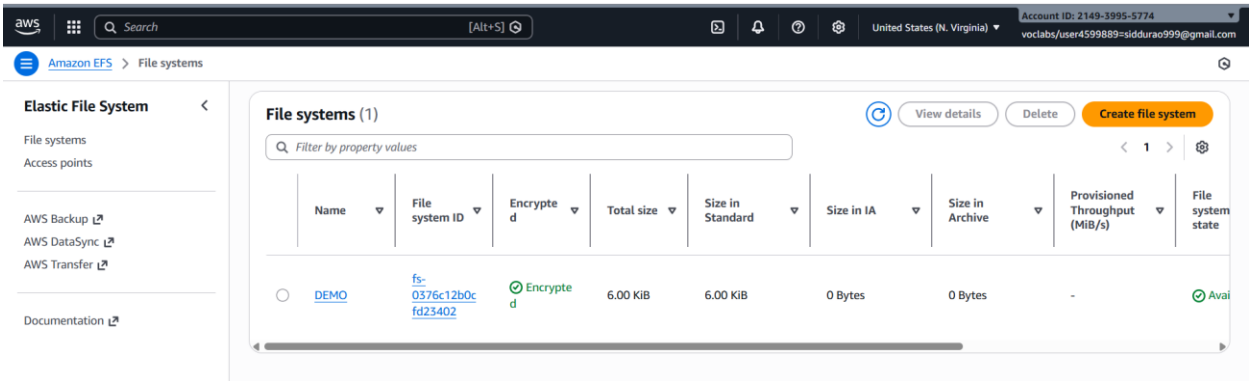
```
[root@ip-172-31-45-9 ec2-user]# yum install -y amazon-efs-utils
Amazon Linux 2023 Kernel Livepatch repository                                                    241 kB/s |  31 kB     00:00
Dependencies resolved.
================================================================================================================================
 Package                        Architecture            Version                         Repository                       Size
================================================================================================================================
Installing:
 amazon-efs-utils               x86_64                  2.4.1-1.amzn2023                amazonlinux                     4.7 M
Installing dependencies:
 stunnel                        x86_64                  5.58-1.amzn2023.0.2             amazonlinux                     156 k

Transaction Summary
================================================================================================================================
Install  2 Packages

Total download size: 4.9 M
Installed size: 10 M
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm                                    3.6 MB/s | 156 kB     00:00
(2/2): amazon-efs-utils-2.4.1-1.amzn2023.x86_64.rpm                               35 MB/s | 4.7 MB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                             28 MB/s | 4.9 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                        1/1
  Installing       : stunnel-5.58-1.amzn2023.0.2.x86_64                                                                     1/2
  Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64                                                                     1/2
  Installing       : amazon-efs-utils-2.4.1-1.amzn2023.x86_64                                                               2/2
  Running scriptlet: amazon-efs-utils-2.4.1-1.amzn2023.x86_64                                                               2/2
  Verifying        : amazon-efs-utils-2.4.1-1.amzn2023.x86_64                                                               1/2
  Verifying        : stunnel-5.58-1.amzn2023.0.2.x86_64                                                                     2/2
================================================================================================================================
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.10.20260202:
    Run the following command to upgrade to 2023.10.20260202:
```

By following these steps, you'll have accessed each EFS-1 instance in separate PowerShell sessions, switched to the root user, created a directory named "efs," installed the Amazon EFS utilities, and listed files in the directory. This setup allows you to configure and manage each instance individually as needed.

Step 4: Attaching EFS to EC2 Instances

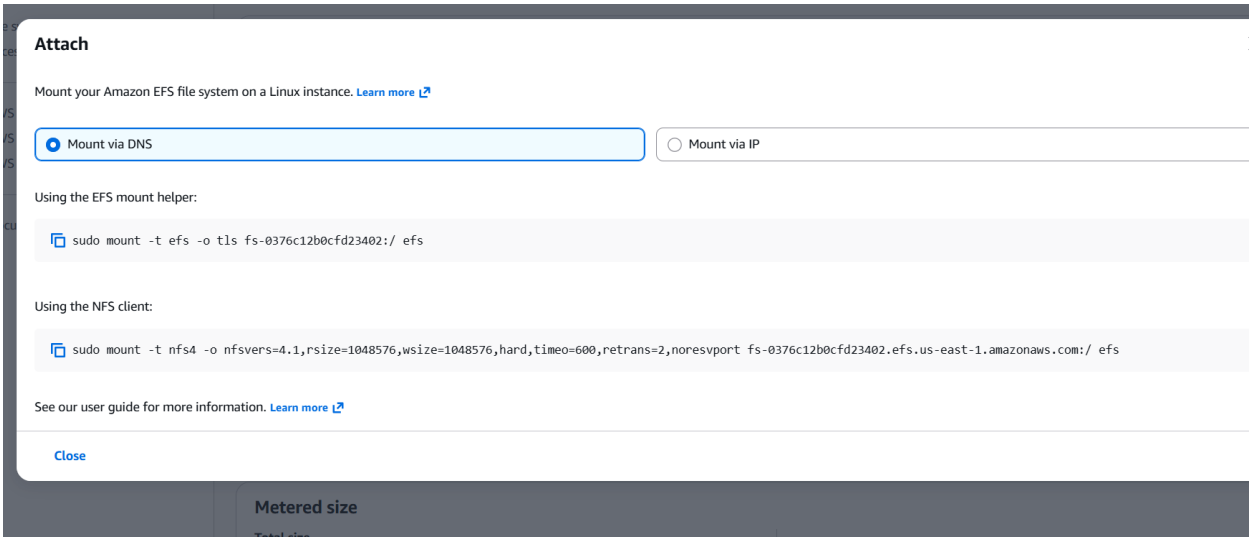
1. Go to the EFS service in the AWS Management Console.

2. Click on the target EFS file system.



3. Click on the "Attach" button.

4. Choose "**Mount via DNS**" option.



5. Copy the displayed command. ,on both ec2 instances(powershell)

6. Paste and execute the copied command in the terminal to mount the EFS file system onto the instance.

```
[root@ip-172-31-0-198 ec2-user]# sudo mount -t efs -o tls fs-06cf881121b15f826:/ efs
```

```
[root@ip-172-31-94-67 ec2-user]# sudo mount -t efs -o tls fs-06cf881121b15f826:/ efs
```

Step 5: Verify and Test EFS

1. Change the directory to EFS on both instances.

2. Create a file on one instance.

```
[root@ip-172-31-0-198 ec2-user]# sudo mount -t efs -o tls fs-06cf881121b15f826:/ efs
[root@ip-172-31-0-198 ec2-user]# cd efs
[root@ip-172-31-0-198 efs]# touch hello.txt
[root@ip-172-31-0-198 efs]# ls
hello.txt
[root@ip-172-31-0-198 efs]#
```

3. Verify that the file automatically synchronizes and appears on the other instance.

```
[root@ip-172-31-94-67 ec2-user]# sudo mount -t efs -o tls fs-06cf881121b15f826:/ efs
[root@ip-172-31-94-67 ec2-user]# cd efs
[root@ip-172-31-94-67 efs]# ls
hello.txt
[root@ip-172-31-94-67 efs]#
```

By above steps, we can successfully create EC2 instances, configured them, created an EFS file system, and attached it to the instances in the same availability zone in the Mumbai region. Now, the instances can seamlessly access and share files stored in the EFS file system.